Hollow Promises? An empirical study of the potential benefits of introducing hollow sections in generative design outcomes.

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Abstract

This study investigates the potential benefits of enabling Generative Design (GD) toolchains to output hollow sections. It is hypothesised that the higher second moment of area of hollow sections would lead to overall stiffer structures for a given mass. Numerical modelling and physical tests were performed on cantilever beams, L-brackets, and MBB beams with solid and hollow sections. This first study showed no significant increase in stiffnesses of the hollow section samples, however physical testing revealed significant load capacity increases of 61.1% and 14.8% for the cantilever and MBB beams respectively but a decrease in load capacity of 11.2% for the L-bracket. A second study was performed to isolate topological effects from manufacturing processes using 3-point bending tests on dimensionally varied tubes. The results from the second study confirmed the hypothesis, demonstrating significant increases in stiffness and load capacity as the second moment of area increased. Together, the results suggest that the impact of hollowing GD artefacts is sensitive to the local geometry, particularly around joint and cross-section transition regions. The findings reveal that hollow geometry only provides structural performance gains in long, straight constant cross-section struts limited by global buckling or flexure. When applied to short, compact structural joints that use material near the neutral axis to resist shear deformation hollowing degraded performance. The paper concludes by formalising design insights and asserting that future GD toolchains must either apply post-process hollowing selectively or integrate natural hole nucleation into generation algorithms themselves.

1 Introduction

Lightweighting refers to reducing a component's mass while maintaining or improving its structural performance. This performance is commonly assessed against two metrics: stiffness, the resistance to elastic deformation under load (typically characterised by the gradient of the load-deflection response); and, load capacity, the maximum load a structure can sustain before failure through yielding, buckling, or fracture. These two metrics are not necessarily correlated: a change in geometry or material distribution that improves one need not improve, or may even degrade, the other.

Generative Design (GD) is an increasingly popular method for designing lightweight components [1]. The advent of cloud computing services has enabled CAD vendors to develop GD toolchains that parallelise the use of computationally intensive Topology Optimisation (TO) algorithms to explore the pareto front of structural design problems. These toolchains enable the designer to maximise the stiffness of a structure for a given weight and over a set of load cases, materials and manufacturing constraints, either by optimising an existing geometry or by generating entirely new designs based upon necessary 'preserve' regions. Using Autodesk Fusion's GD (one of the most widely adopted GD toolchains) academic studies have consistently demonstrated mass savings of >30% when compared to original structures created by expert designers [2, 3, 4, 5]. In the context of GD the objective of lightweighting often refers almost entirely to stiffness since load capacity is often highly computationally costly to predict accurately.

In addition to optimising the distribution of material, a common approach to lightweighting is hollowing/ 'tubularisation', which acts to increase second moment of area. By distributing material further from the neutral axis internal restoring forces are given a larger mechanical advantage to combat external pressures, resulting in higher apparent stiffness and load capacity of the overall structure. This is in contrast to solid structures where the majority of material lies near to the neutral axis [6]. In this way, hollow structures can be designed stiffer and stronger for the same mass; making hollowing an invaluable tool for lightweighting. For example, hollowing a solid round strut via shelling to preserve its mass provides a theoretical stiffness and load capacity increase of 100%.

Like cloud computing, Additive Manufacturing (AM) is an enabling technology for GD. This

is because GD commonly produces biological-looking outcomes that cannot be manufactured via traditional processes [7, 8]. AM similarly provides an enabling role for hollow structures, since traditional manufacturing processes cannot easily create structures with internal voids for complex freeform artefacts. This synergy presents an opportunity to combine the benefits of both lightweighting approaches: GD and hollowing. Despite this opportunity, current commercial GD toolchains typically do not generate hollow outcomes. The reasons for this are three-fold:

1. Algorithmic: GD toolchains are commonly built upon CAD-compatible level set TO approaches which are mathematically incapable of naturally generating hollow structures [9, 10].

2. Cost: Repeated FEA of hollow/thin-walled structures (as required by TO algorithms) is computationally costly, because resolving a thin wall with a volume mesh requires several elements through its thickness, which drives element size, and therefore total element count, across the whole domain [11, 12].

3. Unknown impact: The structural benefits of hollowing are well established for simple prismatic sections, but where hollowing has been evaluated within TO it has been demonstrated only on simple or heavily parameterised geometries [13, 14, 15, 16, 17, 18]. Its impact on the geometrically complex outcomes that GD toolchains produce remains poorly quantified [19, 20].

This paper takes steps towards addressing the third of these reasons. This is achieved by investigating and characterising the impact of hollowing existing GD outcomes. The advantages of hollowing observed in traditional structure design leads to the hypothesis that hollowing GD outcomes will increase their stiffness and strength for a given material mass. The paper continues as follows. Section 2 explores the existing literature to contextualise this work, and establish the gaps that motivate the research questions. Two studies were performed: Study 1 characterises the changes in mechanical performance brought about by hollowing three GD topologies. Study 2 characterises the mechanical performance of simple tubular structures manufactured by the same means as Study 1 to elucidate a mechanism for the observed performance differences in Study 1. Section 3 discusses the methodology of the two studies, including the experimental methods (numerical and

physical) used. Results of the studies are presented in Section 4 and Section 5 respectively. Implications and opportunities associated with the results are then discussed in Section 6 before the paper is concluded in Section 7 with specific attention to GD, AM and future capabilities.

2 Related Work

A key feature of GD toolchains is their integration into CAD programs since it enables streamlined iteration, engineering analysis and outcome postprocessing beyond that offered by standalone TO tools [21, 22]. This integration requires GD toolchains to be built using TO methods that represent geometry in a similar method to standard CAD programs. The majority of modern CAD programs represent 3D components using continuous domain Non-Uniform Rational B-Splines (NURBS) as material boundaries [23]. Material is defined by the region enclosed by NURBS surfaces whilst regions not enclosed define voids. The TO approach most compatible with the smooth, sharp-boundary NURBS approach to geometry definition is the Level Set Method (LSM) [24, 10]. LSM was first developed in [25]. However, it has been shown not to naturally allow for hollow geometry [9, 26]. Any holes that appear in the result must be seeded in the initial structure. Commercial GD toolchains therefore predominantly produce solid structural outcomes. Anecdotal evidence for this lack of ability is presented in Figure 1, where GD outcomes from six of the leading commercial CAD packages across six case studies are demonstrated to be solid.

Additional to their increased stiffness and flexural load capacity, hollow members also have higher resistance to column (Euler) buckling due to their increased radius of gyration for the same cross-sectional area [6]. These advantages lead to hollow structures being exploited in a variety of structural lightweighting applications, for example:

- Space frame design, where tubes are favoured over solid bars due to their increased flexural and torsional stiffness and column buckling resistance [33].
- High-rise tubular building systems, where columns are arranged such that the building acts as a stiff hollow cantilever to resist lateral wind and seismic loads [34].
- Bridge engineering, where box-girders provide the high torsional rigidity required for curved spans and asymmetric lane loading [35].

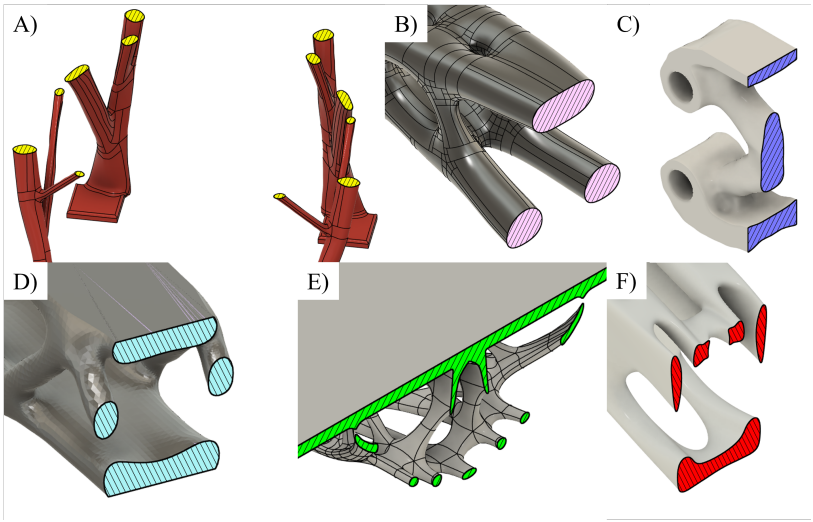

Figure 1: Examples of solid structures generated by a number of leading CAD packages. A) A chair, Autodesk Fusion [27], B) A bottle opener, nTopology [28], C) A bracket, CATIA [29], D) A drone chassis, Siemens NX [30], E) A shelf, Altair Inspire [31], F) A brake pedal, Solidworks [32].

- Wind turbine towers, where hollow tapered cylindrical shells provide maximum bending resistance against aerodynamic forces and decreased column buckling vulnerability [36].
- Drive shafts, where tubes are favoured due to their efficient material distribution in scenarios where there is negligible stress along the axial centre [37].
- Crane booms, where both lattice and telescopic booms leverage hollow geometry to minimise deflection [38].

Hollow geometry has been incorporated into structural optimisation in previous academic studies, albeit in constrained forms. Fiore et al. [16] and Wang et al. [17] optimised the dimensions of parametrised members within predetermined truss and lattice layouts respectively. Bai and Zuo [13] combined internal and external topology description functions to arrange parametrised box sections using Moving Morphable Components (MMC), whilst Zhao et al. [14] obtained hollow members directly by performing a Boolean subtraction of two solid Moving Morphable Bars (MMB), an approach later extended to account for geometrical nonlinearity [39]. Taking different routes, Barberi et al. [40] parametrised the size of holes distributed across a structural plate, and Watson et al. [18] used image processing to replace regions of Solid Isotropic Material with Penalisation (SIMP) outcomes with tubes of conventional engineering cross-section. All of these studies report mass reductions whilst maintaining structural integrity. In each case, however, the starting and solution

layouts are heavily constrained to trusses, lattices, bars, or plates, so the reported benefits are demonstrated on geometry far simpler than those synthesised by commercial GD toolchains.

While the aforementioned applications and methods demonstrate the structural advantages of leveraging simple hollow geometries, there remains little literature exploring the impact of hollowing on highly complex structures such as those produced by GD. One of the few studies in this area attempted to lightweight components by applying an automated shelling operation to a solid GD outcome (Peckham et al. [19]). However, the scope of the study was heavily constrained: the study only considered a single case study and load case, leaving performance under generalisable complex multi-axial stresses untested. Further, while the numerical models predicted significant stiffness gains, the theoretical benefits were not observed during destructive physical testing due to the presence of anisotropy and poor layer adhesion inherent to the Material EXtrusion (MEX) AM process employed.

Thus despite recent studies there remains a lack of rigorous investigation and characterisation of the potential benefits of hollowing in the context of GD. The limitations and experimental challenges faced in recent studies provide important learnings for a more comprehensive characterisation of the theorised benefits and practical considerations. This paper therefore combines numerical simulations alongside destructive physical testing of AM structures. Together they provide both an idealised, theoretical assessment of how altering cross-sections affects structural performance under perfect conditions, and the complex real-world failure modes inherent to manufactured components.

The contributions of this paper are two-fold. First, the quantitative results act as a characterisation of the effects of post-generation hollowing of GD structures in order to address the third reason presented in Section 1. Second, the characterisation results enable establishment of clear design insights into how hollowing should be best applied to complex GD outcome structures.

3 Methodology

This section begins by detailing the methods used in the first study (Section 3.1). It then continues by explaining the investigative processes utilised in the second study (Section 3.2).

3.1 Study 1 Methodology

In this section the methods employed throughout Study 1 to generate, test, and draw comparisons between solid and hollow versions of GD outcomes are explained. Figure 2 splits the study into five stages, which are explained fully in the following subsections.

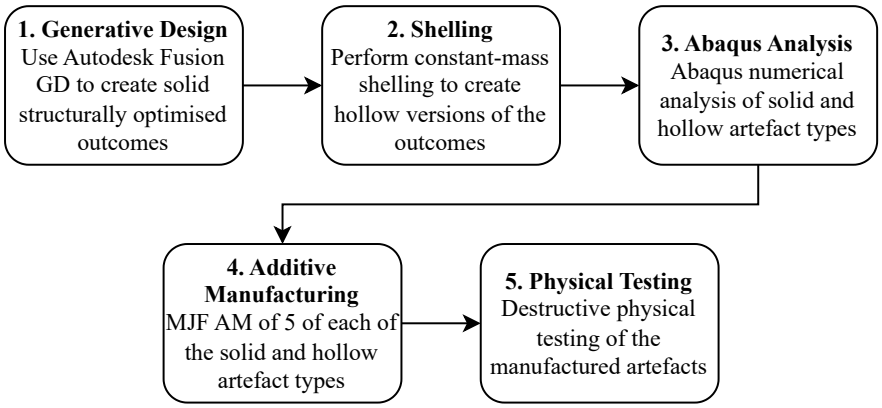

Figure 2: The methodological process taken throughout Study 1. Step X is described in Section 3.1.X of the paper for X=1-5. The results from the testing steps 3 and 5 are presented in Section 4.

3.1.1 Generative Design

To address the first limitation of Peckham et al. [19] concerning generality, three TO benchmark problems were chosen to generate the artefact structures. This ensured that the generated artefacts experienced a range of complex multi-axial stresses across varied topologies. Accordingly, cantilever, L-bracket, and Messerschmitt-Bölkow-Blohm (MBB) beam problems were selected for their variety in boundary conditions, stress states, and resulting topologies [41, 42]. Depictions of these benchmark problems are presented in Figure 3.

Autodesk Fusion GD was used to generate each artefact [43]. The GD problem setups required the common boundary condition regions to be split into multiple regions (e.g. the fixed area in the cantilever problem is split into four regions, each of which is designed to be bolted to a fixed jig) to reduce the volume of the outcomes, keeping AM costs low. All the problems were setup with the objective to maximise stiffness with a safety factor of 1.00 under a load of $0.2kN$. The mass targets set were: cantilever - $60g$, L-bracket - $45g$, MBB - $45g$. No manufacturing constraints were used. Material properties were applied as per the AM process described in Section 3.1.4. After

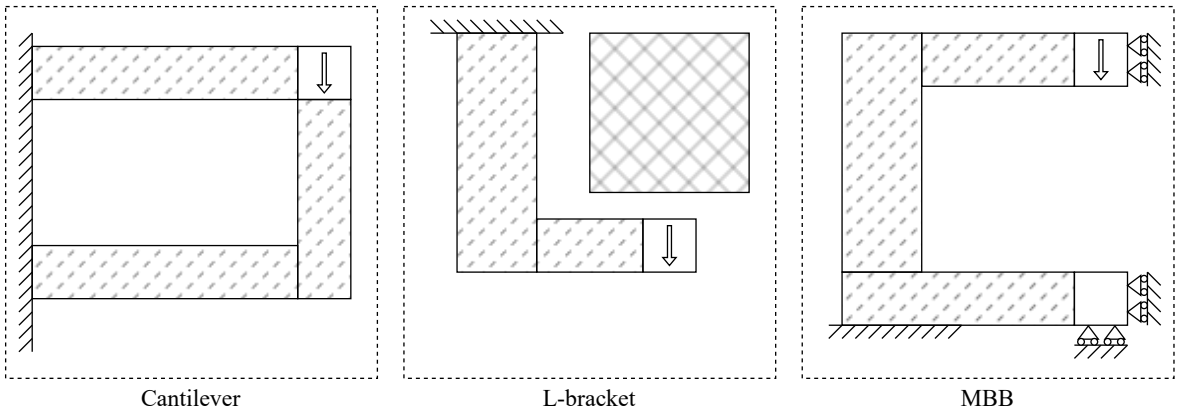

Figure 3: Benchmark TO problem diagrams. Arrows represent regions onto which a load is applied. Un-optimised representative structures are represented using the union of the dashed hatched areas. The cross-hatched area in the L-bracket problem denotes a region where material cannot be placed.

generation the MBB outcome was mirrored to create the full structure. Renders of all the artefact topologies are presented in Figure 4.

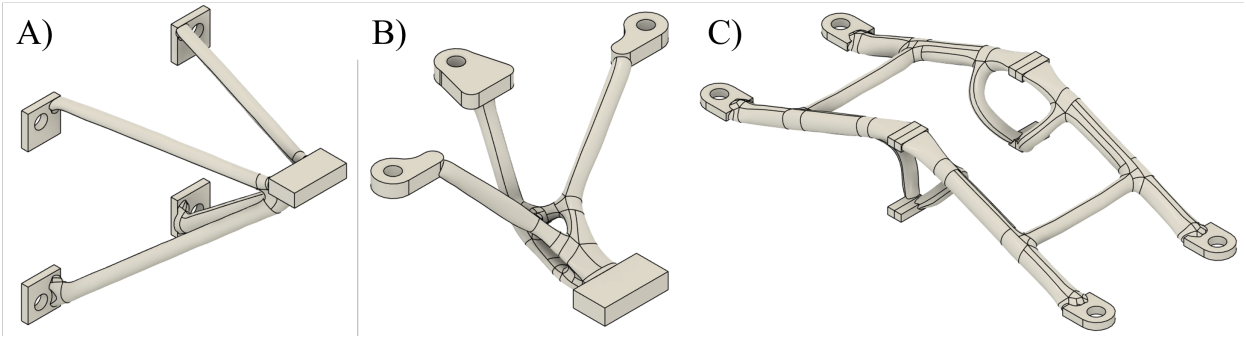

Figure 4: GD outcome for each of the three problems. A) Cantilever, B) L-bracket, C) MBB beam.

3.1.2 Shelling

To hollow the GD outcomes without altering the overall topology and mass of their structure, a constant-mass shelling process was developed [44]. The process iteratively applies different thickness 'outside' shells to the selected region, using a Nelder-Mead optimiser to determine a shell thickness where the solid structure maintains the mass of the solid structure with a tolerance of $\pm 1\%$. An example result of this process is shown in Figure 5 for a cylinder. As can be seen, it is akin to 'flipping' all the material on the inside of the original boundary to the outside, and distributing it evenly around the circumference.

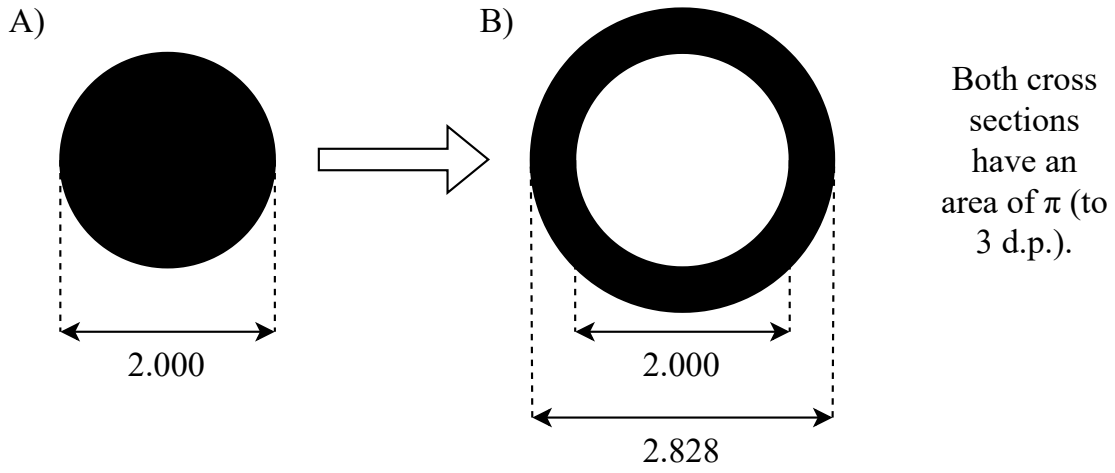

Figure 5: A demonstration of the before (A) and after (B) of constant-mass shelling applied to a solid cross-section to create a hollow cross-section. The method is designed to maintain cross-sectional area.

3.1.3 Structural Analysis

Abaqus was used to numerically predict the stiffness of the solid and hollow artefacts. The CAD models were transferred from Autodesk Fusion as STEP files before using the same Nylon PA11 material properties described in Section 3.1.4. The artefacts were all discretised for analysis using C3D10 mesh elements that enable automatic mesh generation whilst guarding against shear locking. To determine the mesh density required for each artefact, recursive refinement studies were performed with a 1% maximum Von Mises stress tolerance stopping condition. Additionally, the resolution of the hollow artefacts' meshes were increased such that their size was less than a third of the artefact's minimum wall thickness - a rule of thumb recommended by Kim, Jang, and Choi [45] when using solid elements to analyse thin walled structures - if it was not already following the recursive refinement study. The artefacts were loaded with $0.2kN$.

All the artefacts had the same DOF constrained as during their generation. Due to the difficulty of accurately modelling the complex boundary conditions created by bolted fixings, a bounding approach was taken to instead determine a range for the possible stiffness of each structure. This was done by applying constraints in two ways, as displayed in Figure 6. The minimum stiffness case (A) applies linear continuum constraints only on the inside walls of the bolt holes. In the maximum case (B) kinematic constraints are applied to the inside walls of the bolt holes and the

areas under the washers.

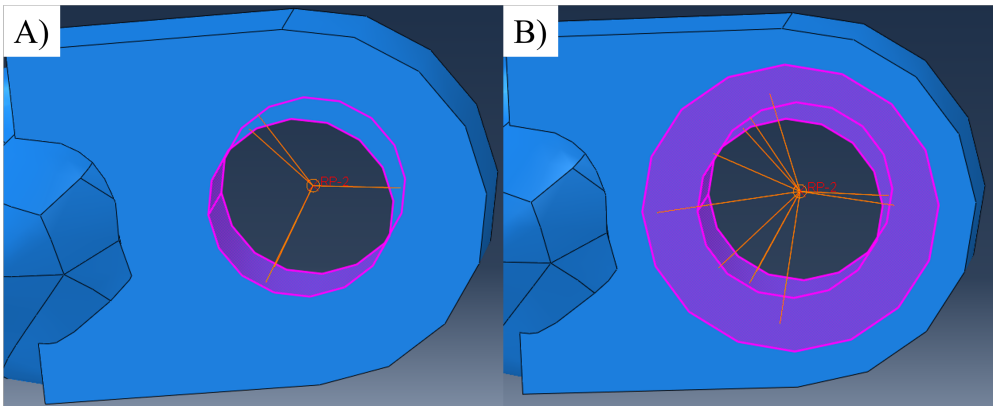

Figure 6: Minimum (A) and maximum (B) stiffness boundary conditions applied at each bolt hole. (A) Linear continuum constraints applied only to the inside wall of the hole. (B) Kinematic coupling constraints applied to the inside wall and the region beneath the washer.

3.1.4 Additive Manufacturing

The use of PLA MEX is a limitation of Peckham et al. [19]. In MEX layers often cool significantly before the nozzle comes back around to deposit more material, leading to reduced stiffness and load capacity in the build direction. Thin walled hollow structures (such as those tested in that study and this) with high surface-area-to-volume ratios are particularly susceptible to increased cooling rates [46]. Furthermore the strength of PLA components is derived from the crystallisation of polymer chains during cooling. When the cooling rate is too rapid full crystallisation cannot occur, leading to reduced mechanical properties [47]. Alongside the weakening caused by these effects, the resulting anisotropy is complicated to model, and cannot be considered during the GD process by current commercial toolchains.

In order to avoid the shortcomings of prior work, it is necessary that the artefacts under consideration are manufactured using processes that produce predictable and modellable isotropic components. This not only enables more accurate performance predictions but also more performant structures since it allows the structure to exploit its material as 'intended' by the GD optimiser. To attain as near as possible isotropic manufactured components, this study leverages Multi-Jet Fusion (MJF) AM. This process offers several advantages for the realisation of complex hollow geometries:

1. **Support-Free Capability:** MJF is a support-free process, allowing for the manufacture of complex, enclosed internal voids without the need for removable support structures. However, the realisation of fully enclosed hollowing requires the inclusion of escape holes to remove unfused powder from the internal design space [48].

2. **High Dimensional Accuracy:** The process maintains high dimensional accuracy (capable of producing clearances as low as $0.2mm$) through the use of detailing agents that generate crisper boundary contours and higher dimensional repeatability than point-based powder bed methods. While the process is capable of producing thin-walled features down to $0.8mm$, the optimal range for structural stability is typically between 1.5-3.0mm [49, 50].

3. **Thermal Stability:** By maintaining a heated powder build volume just below the material melting point, MJF ensures a stable environment with minimal thermal gradients, significantly reducing residual stresses and component warping. This environment acts as a natural annealing chamber, promoting uniform crystallisation throughout the part volume [50].

4. **Isotropy and Predictability:** Unlike point-based energy sources that create transient local thermal histories, MJF utilises a diffuse IR lamp to facilitate a uniform thermal profile across the entire build layer. This results in:

    - Homogenous fusion that provides significantly more isotropic mechanical properties and higher part density than MEX processes [51, 52].
    - Geometry-independent cooling rates in thin-walled regions, ensuring that material properties remain predictable and consistent even as the structure's surface-area-to-volume ratio increases [52].

Five samples of solid and hollow versions of each artefact type were manufactured for a total of thirty samples (fifteen solid, fifteen hollow). Nylon PA11 was used for the MJF process with material properties: density - $1.04g/cm^3$, Young's modulus - $1.76GPa$, Poisson's ratio - 0.42 [53, 54]. All the artefacts of the solid and hollow artefact types of the same topology were printed in the same build direction. To enable powder to escape from the enclosed voids of the hollow versions, $3mm$ diameter escape holes were added in a quantity suggested by a technician from the 3D printing service that was used (Cantilever: 8, L-bracket: 8, MBB: 16). They were placed in

regions that the Abaqus analysis identified as being low stress to minimise their impact on the structures' stiffness and load capacity.

3.1.5 Physical Testing

All the physical tests were performed using an Instron 8872 tensile tester with a $5kN$ load cell. Both the cantilever and L-bracket artefacts were constrained using thick Aluminium jigs whilst the MBB artefacts were bolted directly to the Instron's base plate. Images that demonstrate the setup of each topology are shown in Figure 7. The cantilever and L-bracket artefacts were deflected using a $10mm$ radius ball nose probe. The MBB artefacts were deflected using a $10mm$ thick and $120mm$ wide 'fin' probe designed to apply load evenly onto the two pressure pads. For all artefacts the displacement of the probe was ramped downwards at $0.1mm/s$ until failure or the $25mm$ limit was tripped.

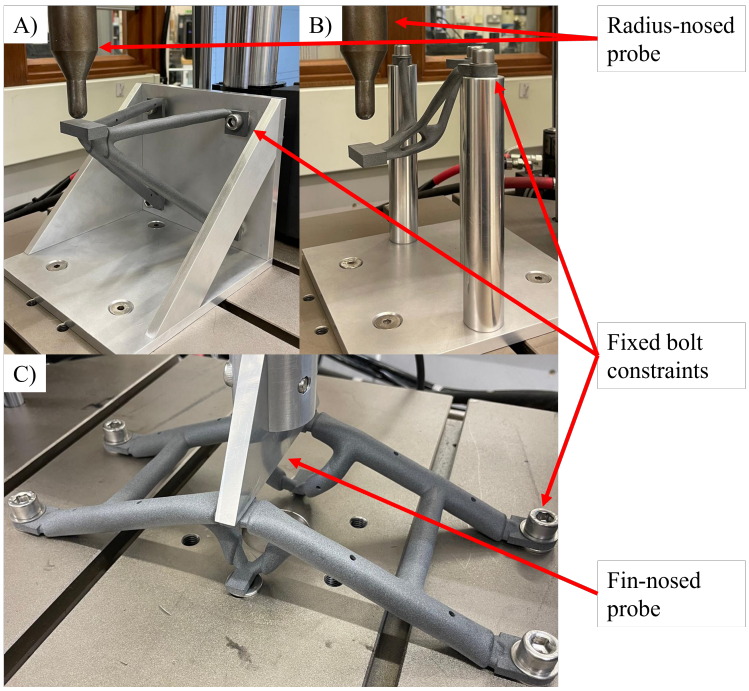

Figure 7: Representative artefact setups in the Instron testing machine. A) Cantilever, B) L-bracket, C) MBB.

3.2 Study 2 Methodology

This section explains the methods employed throughout Study 2 to isolate the purely flexural impact of increasing the second moment of area from the complex topologies and manufacturing effects present in Study 1. Figure 8 splits the process of this investigation into four stages, which are explained fully in the following subsections.

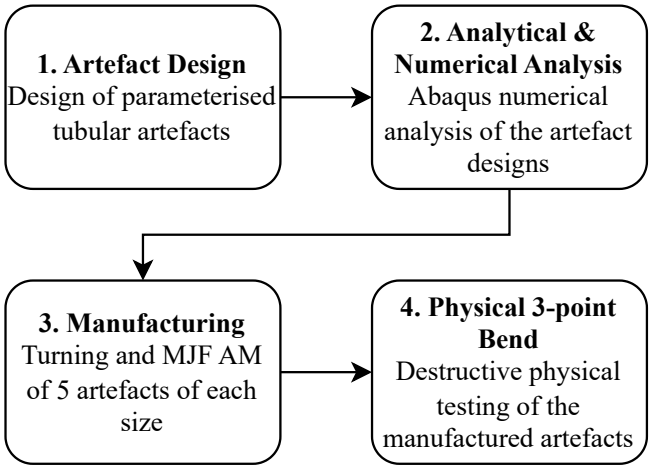

Figure 8: The methodological process taken throughout Study 2. Step Y is described in Section 3.2.Y of the paper for Y=1-4. The results from the testing steps 2 and 4 are presented in Section 5.

3.2.1 Artefact Design

The artefacts for Study 2 were designed to test the core hypothesis that hollowing creates stiffer and stronger structures whilst also isolating the effects of the AM process on the performance of thin-walled structures. To do this simple tubular structures of constant cross-sectional area with no joints were used to test the purely flexural impact of increasing the second moment of area. For a predictable and isotropic baseline against which the MJF AM Nylon PA11 tubes could be assessed, geometrically identical tubes were also manufactured in extruded Aluminium 6063 (Section 3.2.3). If the Nylon PA11 tubes exhibited stiffness and load capacity gains proportional to those of the Aluminium baseline and the analytical/numerical predictions, this would indicate that the MJF AM process is capable of producing performant hollow structures, and that any shortfall observed in Study 1 is attributable to the joint-dominated GD topologies rather than manufacturing limitations.

This study involved a 3-point bend flexure of tubes under conditions that follow ISO178:2019 as closely as possible [55]. The Outer Diameter (OD) and Inner Diameter (ID) of each of the sizes were chosen for manufacturability and such that a near constant cross-sectional area ($176.3 \pm 0.5mm^2$, assuming exact manufacturing - this figure is arbitrary) was maintained between all sizes. Table 1 presents these dimensions. Size A acts as the solid control against which the four increasingly 'hollowed'/ thin-walled sizes B-E are compared. Five artefacts of each size and material were manufactured, for a total of 50 artefacts. ISO178 specifies a 16:1 span-thickness ratio. Since adhering to this is not possible for all the diameters simultaneously, the length of the span was set such that the median size in the range (C) meets this requirement. This gives a span of $16 * 18mm = 288mm$. The standard then specifies that 10% of the span should overhang the pins at each end, yielding artefacts with a total length of $345.6mm$.

Table 1: Tube sizes. A tolerance of $\pm 0.1mm$ was used during the manufacturing process. These dimensions were chosen to keep a constant cross-sectional area of $176.3 \pm 0.5mm^2$.

Size	**Outside Diameter (mm)**	**Inside Diameter (mm)**
A	15.0	0.0
B	15.8	5.0
C	18.0	10.0
D	21.2	15.0
E	25.0	20.0

3.2.2 Analytical & Numerical Analysis

Both analytical and numerical predictions were made for the stiffness of each artefact ahead of physical testing. The numerical models used a C3D8I hex mesh with an elliptical square-to-disc butterfly mesh for the solid rod artefact type (A) and a structured radial mesh for tubular artefact types (B-E). At least 4 elements were used in the radial wall thickness for all the artefact types. Mesh convergence studies were performed for each of the radial, circumferential, and axial dimensions with a stopping condition of 1% Von Mises stress tolerance. The Nylon PA11 properties used were identical to those discussed in Section 3.1.1 and the Aluminium 6063 material properties used were: Young's modulus - $68.9GPa$, Poisson's ratio - 0.33 [56].

The boundary conditions were directly modelled using rigid body rollers of the appropriate dimensions. These rollers interacted with the exterior of the tube via a frictionless contact formu-

lation, which allowed the tube ends to slide freely as it bent. To prevent static singularities, the reference nodes for the support rollers were locked across all six DOF, while the load roller was fixed in all axes except for the vertical Z-axis. Additionally, stabilisation nodes at the top and bottom of the tube were pinned in the X and Y axes to completely stop the tube from rolling and twisting. The stiffness was measured using the deflection resulting from an application of $0.1kN$ at the loading pin.

The deflection was also predicted using Euler-Bernoulli beam theory to explore whether the MJF AM tubes had a larger discrepancy with the theoretical analytical result than the isotropic Aluminium tubes. Timoshenko beam theory, which accounts for shear deformation, unlike Euler-Bernoulli beam theory, was not used because all the tube sizes have a length to thickness aspect ratio greater than 10.

3.2.3 Manufacturing

The Nylon PA11 tubes were manufactured using the same MJF AM process described in Section 3.1.4, with all tubular artefacts printed with their axial direction aligned parallel to the build direction. The Aluminium 6063 tubes were manufactured by turning down the OD of extruded stock with the desired ID of each size, to the same $\pm 0.1mm$ tolerance applied across the artefact set.

3.2.4 Physical 3-point Bend

Physical testing was performed using the same Instron 8872 tensile tester described in Section 3.1.5, under 3-point bend conditions following ISO178:2019 [55]. The supporting pins had a radius of $10mm$ and the loading pin a radius of $25mm$. The loading pin was deflected downwards by the Instron 8872 at a rate of $0.1mm/s$.

4 Study 1 Results

The results from stages 3 & 5 of Study 1's methodology (see Figure 2) are presented in this section. First the numerical predictions are presented before the stiffnesses and load capacities measured using Instron testing are given. Findings that follow directly from the results are flagged here with

a bold, numbered summary statement (F1-F3).

4.1 Numerical Prediction Results

The bounded stiffness predictions for each of the artefact types computed by Abaqus are presented in Table 2. The values demonstrate a trend where the simulation anticipates that hollowing results in a slight stiffness decrease of 11.6% and 8.2% for the cantilever and MBB artefacts respectively. In contrast the L-brackets experienced a 4.0% increase in stiffness attributable to hollowing. In all cases the stiffness change attributable to hollowing is less pronounced than the range created by varied boundary condition application.

Table 2: Bounded predictions for the stiffness of each of the Study 1 topologies. The final column lists the larger-magnitude of the two bound-derived percentage changes; a positive value indicates hollowing increased predicted stiffness.

Artefact	**Min. Stiffness (kN/mm)**	**Max. Stiffness (kN/mm)**	**Max. % Change (Solid vs Hollow)**
Solid Cantilever	0.163	0.268	-11.6
Hollow Cantilever	0.156	0.237	
Solid L-bracket	0.025	0.038	+4.0
Hollow L-bracket	0.026	0.037	
Solid MBB	0.364	0.899	-8.2
Hollow MBB	0.334	0.839	

4.2 Physical Results

F1: Observed stiffness changes due to hollowing are not statistically significant. The load - deflection plots from the destructive physical tests are presented in Figure 9. The mechanical properties of each of the artefact types are summarised in Table 3. As can be seen the observed mean stiffness of each of the artefact types lies within the range of the numerical predictions. These measurements imply a mean stiffness increase of 3.5% for the hollow cantilever artefacts, a difference found to be non-significant using Welch's t-test ($t(7.27) = -0.60$, $p = 0.569$, $Hedges'g = -0.34$). Meanwhile the mean stiffness of the L-bracket artefacts decreased by 7.0% ($t(4.23) = 1.80$, $p =$

$0.142, g = 1.03$), and the MBB artefacts by 14.9% ($t(5.72) = 1.85$, $p = 0.117$, $g = 1.05$); neither of these changes was statistically significant. Of these the numerical analysis only accurately predicted the direction of stiffness change for the MBB artefacts.

F2: Hollowing significantly increased load capacity for the cantilever and MBB topologies, with the MBB result decoupling the two lightweighting metrics within a single artefact. The increases in load capacity of 61.1% and 14.8% between the solid and hollow artefact types of the cantilever and MBB topologies respectively were found to be significant using the same test: for the cantilever ($t(5.18) = -20.15$, $p = 4.06 \times 10^{-6}$, $g = -11.51$); for the MBB artefacts ($t(7.81) = -11.63$, $p = 3.31 \times 10^{-6}$, $g = -6.64$). The MBB result is notable since its stiffness decreased by 14.9% (the largest decrease of the three topologies) simultaneously as its load capacity increased, illustrating within a single artefact that hollowing can be detrimental to stiffness while remaining beneficial to load capacity.

F3: Hollowing resulted in a significant decrease in load capacity for the L-bracket. Conversely to the cantilever and MBB artefacts, the decrease in load capacity of 11.2% for the hollow L-brackets was also found to be significant ($t(5.66) = 4.86$, $p = 0.00329$, $g = 2.78$). This is the only Study 1 topology for which hollowing significantly reduced load capacity, in contrast to the significant increases observed for the cantilever and MBB artefacts (F2).

Table 3: Mean stiffness and load capacity of each Study 1 artefact type measured under Instron testing (five replicates per type). Percentage changes indicate the effects of hollowing each artefact and are calculated relative to the corresponding solid artefact type.

Artefact	**Mean Stiffness (kN/mm)**	**Stiffness SD (kN/mm)**	**Stiffness % Change**	**Mean Load Capacity (kN)**	**Load Cap. SD (kN)**	**Load Cap. % Change**
Solid Cantilever	0.193	0.021	+3.5	0.513	0.013	+61.1
Hollow Cantilever	0.200	0.015		0.826	0.032	
Solid L-bracket	0.031	0.003	-7.0	0.453	0.021	-11.2
Hollow L-bracket	0.029	0.000		0.403	0.010	
Solid MBB	0.602	0.047	-14.9	2.436	0.053	+14.8
Hollow MBB	0.512	0.098		2.796	0.045	

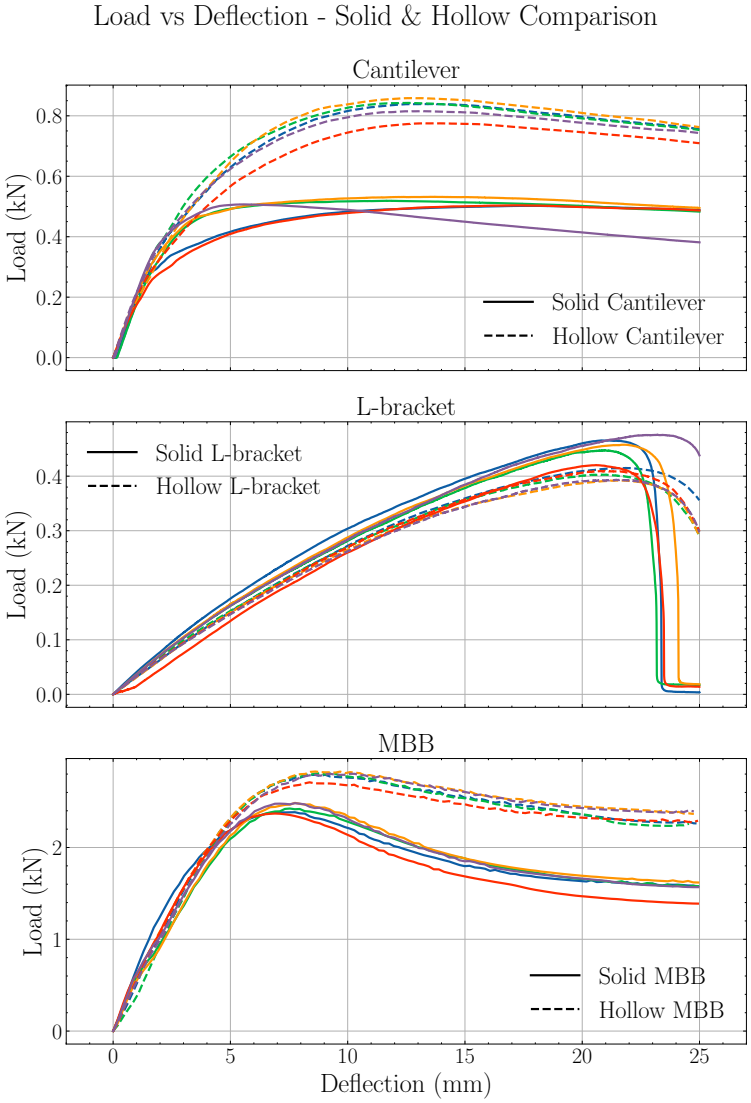

Figure 9: Load-deflection curves from destructive Instron testing of the three Study 1 artefact topologies, comparing solid (solid lines) and hollow (dashed lines) versions. Five replicates are shown per artefact type. Note that the load axis scales differ between the three subplots.

4.3 Manufacturing Validation

Prior to destructive testing the artefacts were weighed for validation of the manufacturing process. Table 4 presents the masses of the artefacts. For all three topologies, there is a statistically significant mass loss between the solid and hollow versions using Welch's t-test, with mean reductions of 1.8% for the cantilever ($t(7.87) = 4.97$, $p = 0.00114$, $Hedges'g = 2.84$); 3.2% for the L-bracket ($t(4.47) = 2.89$, $p = 0.0391$, $g = 1.65$); and 4.1% for the MBB artefacts ($t(5.32) = 12.82$, $p = 3.36 \times 10^{-5}$, $g = 7.33$). This loss is expected, since material was removed from the hollow artefacts to create the escape holes required to release unfused powder (Section 3.1.4). The magnitude of the loss loosely tracks the number of escape holes introduced into each topology (8, 8, and 16 for the cantilever, L-bracket, and MBB respectively): the MBB artefacts, which required the most holes, exhibited the largest mean loss, though the L-bracket's loss exceeds that of the cantilever despite an equal hole count, demonstrating that local wall thickness and hole placement also influence the amount of material removed.

Table 4: Manufactured artefact masses in grams (five replicates per type). The final row gives the mean percentage mass change of the hollow artefacts relative to the solid artefacts of the same topology, attributable to material removed for the escape holes.

Artefact	**Cantilever**		**L-bracket**		**MBB**	
#	**Solid**	**Hollow**	**Solid**	**Hollow**	**Solid**	**Hollow**
1	60.7	59.6	43.3	40.7	87.6	84.4
2	61.4	59.5	42.8	40.4	87.0	83.2
3	60.9	60.3	43.1	42.4	87.2	83.1
4	61.1	60.2	43.3	42.5	87.5	83.7
5	60.6	59.7	42.8	42.4	87.3	84.2
Mean	60.9	59.9	43.1	41.7	87.3	83.7
Std Dev	0.32	0.36	0.25	1.04	0.24	0.58
Mean % Change	-1.8		-3.2		-4.1	

After destructive physical testing, cross-sections were taken from each of the hollow artefacts for wall thickness measurement. The cross-sections were resin potted and then digital calipers applied in six locations using a microscope. Figure 10 illustrates how these measurements were taken. The target and mean wall thicknesses for each of the topologies are presented in Table 5. The hollow artefacts were consistently manufactured with $\geq 0.2mm$ thinner walls than intended. This reveals inaccuracies in the MJF AM process that prevent the artefacts from being manufactured as large

and stiff as designed. This systematic manufacturing error also contributes to the explanation for the lower masses observed in the hollow artefacts.

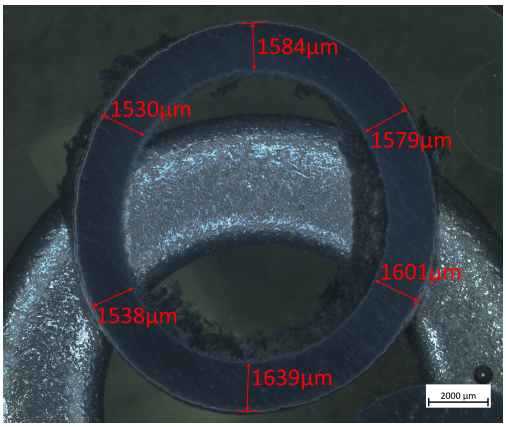

Figure 10: Representative resin-potted cross-section of a hollow Study 1 artefact (Hollow Cantilever shown), annotated with the six wall-thickness measurements taken around the circumference using a Zeiss microscope.

Table 5: Target versus mean measured wall thickness of the hollow Study 1 artefacts, from six microscope measurements per cross-section. Bias is the mean measured thickness minus the target thickness; a negative value indicates walls manufactured thinner than intended.

Artefact	**Target (mm)**	**Mean (mm)**	**Std Dev (mm)**	**Bias (mm)**
Hollow Cantilever	1.756	1.555	0.039	-0.201
Hollow L-bracket	1.700	1.486	0.046	-0.214
Hollow MBB	2.100	1.893	0.045	-0.207

Together, these results indicate that globally increasing the second moment of area of the structures' cross-sections via constant-mass shelling did not yield a reliable stiffness benefit for these complex topologies, consistent with the findings of other studies [19, 20]. However in the case of the cantilever and MBB artefacts the load capacity was significantly increased.

5 Study 2 Results

The sensitivity of the numerical stiffness bounds to the assumed boundary conditions (Section 4), together with the thinner-than-intended walls revealed by microscopy above (a deviation not accounted for in the numerical models) leaves it unclear whether this outcome reflects the joint-dominated GD topologies themselves or a limitation of the MJF AM process. Study 2 was designed

to resolve this ambiguity using simplified tubular geometry, free of both complex joints and with a comparison against non-AM control artefacts. As in Section 4, the finding that follows directly from these results is flagged here with a bold, numbered summary statement (F4).

5.1 Analytical & Numerical Prediction Results

The Euler-Bernoulli and numerical predictions for the stiffness of the tubes are presented in Table 6. Both methods predict monotonic increase in stiffness with increasing hollowness (i.e. with increasing outer diameter at constant cross-sectional area), confirming the theoretical hypothesis introduced in Section 1. Relative to the solid control (size A), the Euler-Bernoulli and numerical models respectively predict stiffness increases of up to 355.6% and 309.7% by size E for the Aluminium 6063 tubes, and 354.6% and 308.4% for the Nylon PA11 tubes. These gains substantially exceed the 100% theoretical benchmark associated with a simple solid-to-hollow constant mass outside shelling conversion, reflecting the more aggressive geometric change applied progressively across the full A-E size range.

The two prediction methods match with a high degree of accuracy for the more solid sizes for both materials, however the discrepancy between them grows systematically as OD increases and wall thickness decreases. For size A, the discrepancy was 1.5% and 2.3% for the Aluminium 6063 and Nylon PA11 respectively, rising to 2.3% and 2.9% for size B, 3.4% and 4.4% for size C, 5.9% and 7.0% for size D, and reaching 11.4% and 12.2% for size E. This near-monotonic growth in discrepancy with increasing hollowness, observed consistently across both materials, indicates a systematic effect of wall thickness on prediction agreement rather than a material-specific numerical artefact. Both prediction methods hypothesised stiffness increases with increased 'hollowing', though their agreement with one another degrades in the thin-walled regime where the theoretical benefits of hollowing are greatest.

5.2 Physical 3-point Bend Results

F4: Tests on tubes confirmed significant increases in stiffness and load capacity of hollow tubes are observed as the second moment of area increases. Table 7 presents the mean measured stiffness and load capacity of all the tube sizes after physical testing. Both metrics increased monotonically with the second moment of area for both materials. For the Aluminium

Table 6: Stiffness predictions for the tubular control artefacts, comparing analytical Euler-Bernoulli beam theory calculations against numerical Abaqus models. Percentage changes are relative to the solid size A. Size A: [OD = 15.0mm, ID = 0.0mm], Size B: [OD = 15.8mm, ID = 5.0mm], Size C: [OD = 18.0mm, ID = 10.0mm], Size D: [OD = 21.2mm, ID = 15.0mm], Size E: [OD = 25.0mm, ID = 20.0mm].

Material	**Size**	**Euler-Bernoulli Stiffness (kN/mm)**	**E-B % Change**	**Numerical Stiffness (kN/mm)**	**Num. % Change**
Aluminium 6063	A	0.3440	-	0.3388	-
	B	0.4193	+21.9	0.4099	+21.0
	C	0.6455	+87.7	0.6233	+84.0
	D	1.0287	+199.0	0.9676	+185.6
	E	1.5673	+355.6	1.3884	+309.7
Nylon PA11	A	0.0088	-	0.0086	-
	B	0.0107	+21.6	0.0104	+20.8
	C	0.0165	+87.5	0.0158	+83.4
	D	0.0263	+198.9	0.0245	+184.6
	E	0.0400	+354.6	0.0351	+308.4

6063 tubes, mean stiffness rose by +251.3% from the solid control (size A) size E, while mean load capacity rose by +67.3%. The Nylon PA11 tubes exhibited proportionally larger gains over the same size range: mean stiffness increased by +372.2%, and mean load capacity by +47.0%. For both materials, the relative gain in load capacity was consistently smaller than the corresponding relative gain in stiffness at each size, consistent with the onset of local shell buckling increasingly limiting the achievable load in the thinnest-walled sizes. The Nylon PA11 tubes showed a larger relative stiffness gain than the Aluminium 6063 tubes despite their lower absolute stiffness. These results confirm the hypothesis that stiffness and load capacity increase with the second moment of area, even for MJF AM manufactured artefacts.

Figure 11 depicts how the predictions in Table 6 overestimated the measured stiffness in every case, with no underpredictions observed across either material or any tube size. For Aluminium 6063, the overprediction is broadly monotonic across the size range rather than confined to the thinnest walled sizes: the Euler-Bernoulli and numerical predictions overestimate stiffness by +14.6% and +12.9% for size A, easing marginally to +14.0% and +11.4% for size B, before rising steadily to +20.5% and +16.3% for size C, +33.7% and +25.8% for size D, and +48.6% and +31.6% for size E. For Nylon PA11 the overprediction is substantially larger throughout: the

analytical predictions range from +61.9% to +65.4% for sizes A-C before reducing to +44.7% and +55.9% for sizes D and E, while the numerical predictions follow a similar pattern at +57.6% to +60.5% for A-C and +34.7% and +36.8% for D and E. Unlike the GD outcomes, no systematic thinner-than-intended wall thickness was observed in the MJF AM tubes as confirmed using caliper measurements of crosssections following destructive testing.

Table 7: Mean measured stiffness and load capacity from the 3-point bend tests of the tubular control artefacts (five replicates per size and material). Percentage changes are relative to the solid control (size A) of the same material. Size A: [OD = 15.0mm, ID = 0.0mm], Size B: [OD = 15.8mm, ID = 5.0mm], Size C: [OD = 18.0mm, ID = 10.0mm], Size D: [OD = 21.2mm, ID = 15.0mm], Size E: [OD = 25.0mm, ID = 20.0mm].

Material	**Size**	**Mean Stiffness (kN/mm)**	**Stiffness % Change**	**Mean Load Capacity (kN)**	**Load Cap. % Change**
Aluminium 6063	A	0.3003	-	2.4667	-
	B	0.3678	+22.5	2.6252	+6.4
	C	0.5358	+78.5	2.7167	+10.1
	D	0.7694	+156.2	3.4437	+39.6
	E	1.0549	+251.3	4.1262	+67.3
Nylon PA11	A	0.0054	-	0.1684	-
	B	0.0065	+19.1	0.1901	+12.9
	C	0.0100	+84.1	0.2576	+53.0
	D	0.0182	+234.4	0.3905	+131.9
	E	0.0257	+372.2	0.4160	+147.0

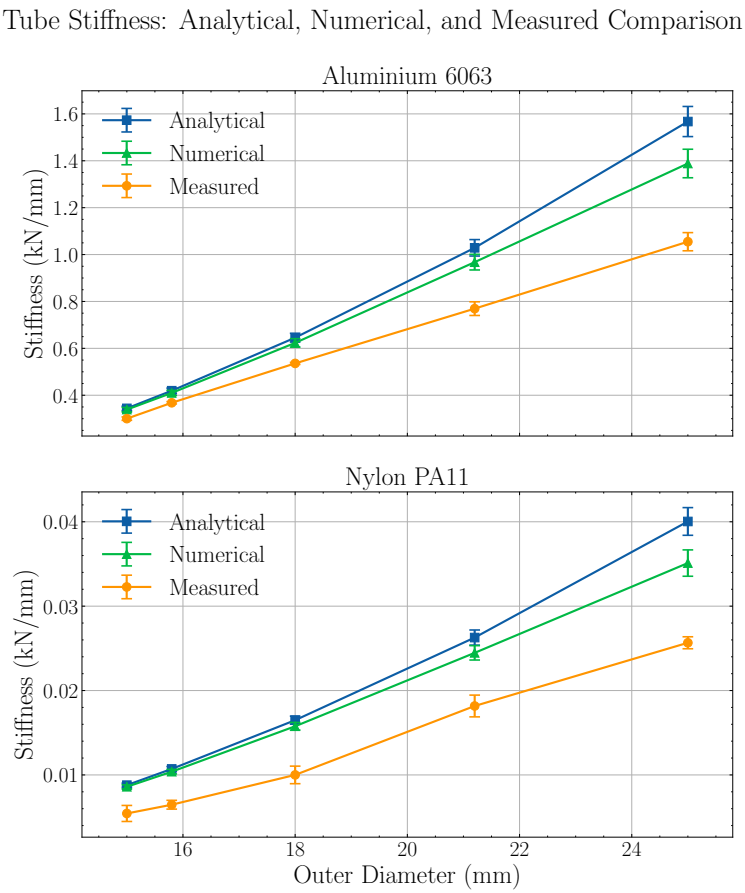

Figure 11: Comparison of Euler-Bernoulli analytical, numerical (Abaqus), and measured mean stiffness for each tube size and material. Error bars on the Analytical and Numerical series propagate the $\pm 0.1mm$ diameter manufacturing tolerance; the Measured series instead shows one standard deviation across the five physical replicates.

6 Discussion

This paper makes two contributions. First, characterising and quantifying the structural impact of hollowing GD outcomes. Second, providing design insights derived from this characterisation. To clearly delineate these contributions, Section 6.1 focusses on structural characterisation, exploring stiffness, load capacity, and modelling accuracy. Subsequently, Section 6.2 translates these characterisation findings into concrete design insights. Findings that follow directly from the results (F1-F4) are flagged in Sections 4 and 5 themselves. The remaining, more explanatory or emergent findings are flagged within Section 6.1. Section 6.2 then refers back to all of these findings by number as it builds design insights upon them.

6.1 Structural Characterisation

This section discusses the characterisation of the structural impact of hollowing GD outcomes, beginning with its effect on stiffness in Section 6.1.1 and load capacity in Section 6.1.2 within the joint-dominated Study 1 topologies. Throughout the section discussion focusses on different structure regions, Figure 12 is provided for reference. Section 6.1.3 then discusses the isolation of these effects from the manufacturing process using the controlled tube geometries in Study 2, before Section 6.1.4 assesses the accuracy of the numerical models underpinning both studies.

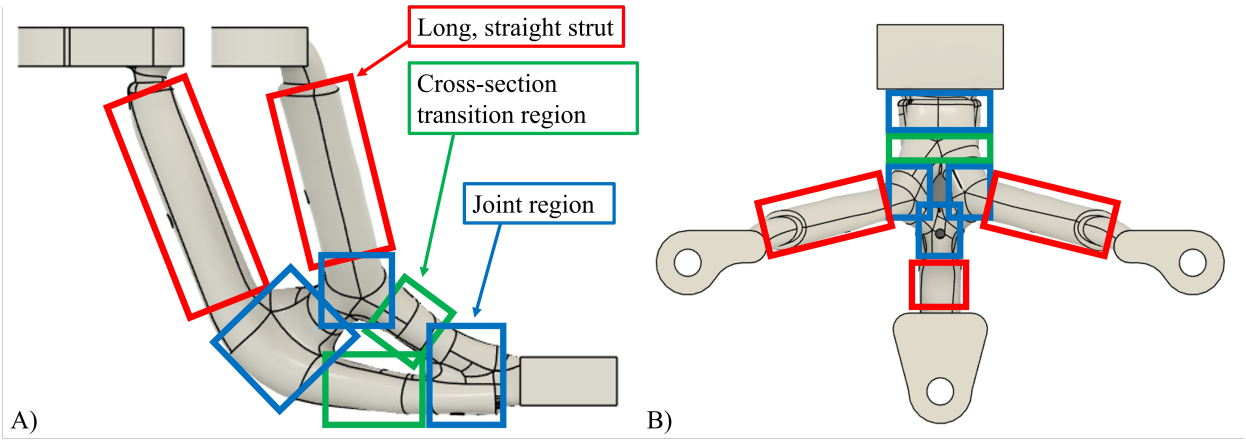

Figure 12: Region types discussed within Section 6. The red boxed regions are long, straight struts, green boxed regions are where the cross-section is transitioning between e.g. circular and elliptical, and blue boxed regions are joints. A side view (A) and top view (B) of the hollowed version of the L-bracket artefact are used to illustrate the regions.

6.1.1 Stiffness in Joint-Dominated Topologies

As reported in Section 4 (F1), the numerical models and physical tests of the Study 1 artefacts (cantilever, L-bracket, MBB) demonstrate that hollowing via constant-mass shelling does not yield a statistically significant stiffness advantage. The physically measured effect was not only small but transitioned from positive and negative, with mean stiffness rising 3.5% for the hollow cantilever but falling 7.0% and 14.9% for the L-bracket and MBB respectively, and none of these relative changes were found to be statistically significant. Numerical models utilising nominal geometries were similarly inconsistent, predicting a stiffness decrease for the cantilever (-11.6%) and MBB (-8.2%) artefacts but a modest increase for the L-bracket (+4.0%), such that the direction of the physically observed change was only correctly predicted for the MBB topology. This disagreement is compounded by the sensitivity of the numerical predictions to the assumed boundary conditions, which for the MBB topology bounded a stiffness range of approximately 147-151%, over an order of magnitude larger than the 4.0-11.6% change attributable to hollowing in any topology. Also relevant is the fact that the numerical analysis used nominal, as designed geometry, whereas microscopy later revealed the manufactured hollow artefacts had walls $\geq 0.2mm$ thinner than intended, a deviation never accounted for during the modelling.

F5: The observed inconsistency in stiffness responses arises from sensitivity to local geometry, particularly around joint and cross-section transition regions. A second study performed to isolate design and process variables revealed that the results arise as a consequence of the complex internal load paths of the GD topologies. Many of the generated structures contain short, compact regions at the joints between struts, particularly the L-brackets. In these regions, shear deformation becomes a significant proportion of the overall deflection. Since material near the neutral axis is required to resist heavy shear loads, applying a hollowing operation to these joints removes critical material, decreasing deformation resistance and negating the theoretical advantages of a higher second moment of area in these regions. The conclusion is drawn that the small, topology dependent net stiffness changes observed here are consistent with this joint mechanism acting alongside the offsetting benefit of hollowing in 'strut like' regions, rather than with a uniform stiffness penalty across the whole structure. Beyond the joints themselves, Figure 12 also identifies a third region type: the short transition zones where the cross-section is actively changing shape (e.g.

from circular to elliptical, or flattening/swaging), typically flanking the joints. These transitions are neither simple struts nor discrete joints, and shelling a geometry that is already changing shape may compound local stress concentrations at exactly the section where the second moment of area is least well-defined, offering a further, geometry-specific explanation for the inconsistent stiffness response. The results presented in this paper do not make it clear whether the rate of change of cross-section has any impact on the overall structure's stiffness, further investigation is required.

6.1.2 Topology Dependent Load Capacity and Buckling Behaviour

As reported in Section 4 (F2), despite the lack of a reliable stiffness benefit, hollowing significantly impacted the load capacity of the Study 1 artefacts, although the effect was highly dependent on the topology of the structures and their corresponding failure modes. For the cantilever and MBB beam topologies, hollowing resulted in significant load capacity increases of 61.1% and 14.8%, respectively. This is due to these specific topologies incorporating long, straight struts that are placed in compression and dominated by global Euler buckling. In these regions, distributing material further from the neutral axis increases the radius of gyration and therefore buckling resistance. This directly illustrates the decoupling of the two lightweighting metrics motivating this study.

As reported in Section 4 (F3), the hollow L-bracket artefacts demonstrated a significant 11.2% decrease in load capacity. This artefact features short, compact, and complex angled joints that experience multi-axial stresses. This topology predisposes the structure to local shell buckling, an effect that was further exacerbated by the $\geq 0.2mm$ thinner (up to 13.6% reduced) than intended walls produced during the MJF AM process observed in Section 4.3. Although the wall thickness deficit was of a broadly similar relative magnitude across all three topologies, local buckling of the flat joint regions is expected to be considerably more sensitive to small thickness reductions than the global column buckling of the slender struts, which may explain why only the L-bracket's load capacity was measurably reduced. This is consistent with the reduced load capacity gains observed for the thinnest walled tubes in Study 2, indicative of the onset of local shell buckling at large deflections in these sizes also. The L-bracket's joints are also flanked by pronounced cross-section transitions, where the changing section geometry may locally concentrate the wall-thickness deficit identified in Section 4.3, compounding the local shell buckling sensitivity discussed above.

6.1.3 Decoupling Geometry from Manufacturing: Study 2

As reported in Section 5 (F4), to separate the flexural impact of an increased second moment of area from the geometric complexity of the GD outcomes, Study 2 was conducted using rod and tubular artefacts. These control tests verify that the AM process is capable of producing performant thin walled structures, and indeed that the MJF AM Nylon artefacts experienced larger stiffness increases as their walls became thinner in comparison to the extruded Aluminium artefacts. Notably, despite their lower absolute stiffness values, the Nylon PA11 tubes exhibited larger relative gains in both stiffness and load capacity than the Aluminium 6063 tubes, indicating that the structural benefit of an increased second moment of area is not diminished, and may even be amplified, for lower modulus AM materials. This is a favourable finding for GD toolchains, since the majority of AM processes currently used to realise their outcomes, including the MJF process used throughout this study, produce polymer rather than metal components. This suggests that the failure of the hollowing hypothesis in the outcomes of Study 1 arises from their short, shear-dominated joint regions, and the adjacent transition zones where the material cross-section changes from circular to elliptical or flattened (swaged) and vice versa, rather than manufacturing limitations of the MJF process.

F6: Increases in load capacity of tubes consistently lags behind stiffness gains as hollowness increases, the inverse of the decoupling was observed in joint-dominated topologies. Load capacity gains, however, consistently lagged behind stiffness gains at every size for both materials. This is most marked for the Nylon PA11 artefacts in Study 2, where stiffness rose 372.2% against a load capacity rise of only 147.0% as the tubes became increasingly susceptible to local shell buckling at large deflections, capping the achievable load even as stiffness continued to climb with increasing hollowness. This is the inverse of the pattern observed in the joint dominated Study 1 artefacts, where hollowing left stiffness largely unchanged while load capacity rose substantially, indicating that the two metrics decouple in opposite directions depending on whether a structure is limited by global buckling/ flexure or by local shear at its joints. While this study validates the strut/ flexure half of this mechanism through controlled testing, the joint/ shear half, and the intermediate transition regions where section geometry is itself changing remain supported only by inference from the Study 1 geometries. A short span 3-point bend or direct

shear test on short, thick walled tubes, potentially incorporating a controlled circular-to-elliptical transition, would be a natural extension to test this hypothesis but remains future work.

6.1.4 Accuracy and Limitations of the Numerical Models

The numerical modelling approach was successful in determining the relative changes in stiffness between the Study 1 solid and hollow structures, and the minimum and maximum stiffness bounding conditions contained the mean physically measured stiffness metrics of the tested artefacts. However, as discussed in Section 6.1.1, this bounding range itself spans as much as 147-151% for the MBB topology, so containment within such wide bounds is a weak test of the model's predictive accuracy, and offers limited confidence in the absolute stiffness predictions for these geometrically complex, joint dominated topologies.

F7: Modelling consistently overestimated the measured tube stiffness, with the overprediction growing as wall thickness decreased. Study 2 offers a more controlled assessment of the numerical and analytical models, free of the joint complexity and wide boundary condition uncertainty present in the Study 1 artefacts. As reported in Section 5, both the Euler-Bernoulli and numerical predictions overestimated measured stiffness for every size and material, with the overprediction generally growing as wall thickness decreased. This is consistent with Euler-Bernoulli beam theory's neglect of shear deformation, which becomes increasingly significant as the wall thins, and indicates that the numerical model is subject to similar thin wall effects despite its more complete representation of the geometry. However, this shear deformation effect alone cannot fully account for the overprediction already evident for the tubes approaching a solid cross-section, where wall thickness effects are minimal: the discrepancy already exceeds +14% for Aluminium 6063 and +55% for Nylon PA11 at size A for both methods, which points instead to the assumed material properties. The magnitude and persistence of this offset suggests that the assumed Young's modulus of $1.76GPa$ overstates the stiffness of the as printed Nylon PA11 parts, or that there is significant compliance in the test fixture at the low load levels these specimens require (however no compliance of this type was observed during testing).

This overestimation has direct influence on how the Study 1 artefact predictions should be interpreted, since the Study 1 models relied on this same Nylon PA11 modulus assumption. Because the Study 1 comparisons are relative (solid versus hollow within a topology), an error that

inflates both predictions by a similar factor would be expected to largely cancel out of the reported percentage changes, limiting its impact on Study 1's conclusions. This cannot be confirmed from the present data, however. The measured Study 1 stiffnesses fell within the wide min-max bounds produced by the boundary condition uncertainty discussed above, but this containment is equally consistent with the modulus overestimation and boundary condition uncertainty coincidentally offsetting one another as it is with both sources of numerical error being independently well characterised. Separately, neither study numerically modelled ultimate load capacity via plastic deformation or buckling. Capturing these complex, nonlinear failure mechanisms for anisotropic AM materials under multi-axial stress states fell outside the scope of the stiffness-focused numerical analysis performed in this paper.

6.2 Design Insights and Implications for GD Software

Building upon the structural characterisation detailed in the preceding section, this section attempts to form design insights from the observed results. Together, the structural results indicate that the value of hollowing to a GD toolchain depends not only on where material is removed from, but on when that decision is made relative to the GD process itself. The post process hollowing tested in this study modifies a topology that was generated on the assumption of a fully solid material distribution: the optimiser had no way to anticipate that its strut and joint regions would later be hollowed, and so had no opportunity to reinforce shear critical joints or otherwise adapt the solid topology in anticipation of this. This may explain why a single, globally applied hollowing operation produced such inconsistent outcomes (F1 and F5), helping load capacity in slender, buckling governed regions while harming it at shear dominated joints (F2 and F3), and, as the MBB result shows, doing so while moving stiffness and load capacity in opposite directions within the same artefact (F2).

Two distinct routes toward leveraging hollow geometry more effectively follow from this. The first is to make post process hollowing selective rather than global, restricting it to regions where it is expected to help. This would require a means of distinguishing 'joint like' and 'transition' from 'strut like' regions within an existing solid topology (F5) and applying hollowing only where a region is governed by flexure or buckling. Either a geometric measure, such as local strut slenderness, or a mechanical measure, such as the local ratio of shear to bending strain energy may appropriate.

This route requires no change to the underlying optimisation algorithm and could be implemented as a refinement of the constant-mass shelling process used here. However, it is still applied after the fact to a topology that was optimised without any anticipation of subsequent hollowing, and so cannot exploit the second moment of area benefit.

The second route is to make the optimiser capable, during topology refinement, of naturally nucleating hollow voids at regions in the structure where there is incentive to do so. This would allow the optimiser to allocate solid, shear resistant material at anticipated joint locations while permitting, or actively favouring, hollow cross sections in the slender regions between them, rather than generating a solid optimised topology and retrospectively applying hollowness to it. This directly addresses the algorithmic barrier identified in Section 1: existing level set TO methods cannot natively nucleate holes, and so cannot make this kind of concurrent decision. Whichever route is taken, restricting hollowing to a subset of the domain would also ease the cost barrier, since only that subset would then require the fine meshing that thin walled features demand for accurate FEA.

This second route also inherits the accuracy limitations identified in Section 6.1.4 (F7). Both the Euler-Bernoulli and numerical predictions in Study 2 grew progressively less accurate as wall thickness decreased, with the overprediction of stiffness worsening most for the thinnest walled, most heavily hollowed sizes, precisely the regime in which hollowing is expected to deliver the largest structural benefit. An optimiser that evaluates candidate hollow geometries via repeated FEA during generation would therefore be relying most heavily on its least accurate predictions when deciding how aggressively to hollow a given region, risking either overly conservative outcomes or unanticipated in service underperformance. Before hollow aware GD can be deployed with confidence, this thin wall prediction error would need to be characterised and, where possible, corrected for (for example via a wall thickness dependent modulus correction of the kind implied by the Study 2 results) rather than assumed to be uniform across the design space as is presently the case for solid outcomes.

Finally, the buckling sensitivity result discussed in Section 6.1.2 (F3) suggests that either route should account for manufacturing realistic wall thicknesses rather than nominal design geometry alone, since a thickness deficit that is inconsequential for a slender strut may be a key factor in whether a joint prematurely fails. More broadly, this paper provides an initial, quantified basis

across three topologies for where the benefits and risks of hollowing lie. Hollowing left stiffness statistically unchanged, with non-significant changes ranging from +3.5% to -14.9% across the three topologies (F1). In contrast, it drove significant load capacity swings: up to +61.1% at slender, buckling-governed struts (F2), but -11.2% at shear-critical joints (F3). This shows that the primary risk of naive post process hollowing is a loss of load capacity at identifiable joint regions rather than a general loss of stiffness, a finding that should inform which of the two aforementioned routes is prioritised as GD toolchains begin to incorporate hollow geometry.

7 Conclusion & Future Work

This paper provides two contributions. The first concerns the characterisation of the structural impact of post-generation hollowing on GD outcomes. The second relates to actionable design insights for GD methods. Addressing the first contribution, numerical and physical testing of 30 AM components demonstrated that global hollowing of cantilever, L-bracket, and MBB beam topologies yielded no significant stiffness improvements for the cases and AM process considered. However, hollowing heavily influenced load capacity based on topology: cantilever and MBB artefacts saw increases of 61.1% and 14.8%, while the L-bracket's capacity decreased by 11.2%. A second study using constant cross-section tubes confirmed these findings were due to topological complexity of the artefacts rather than manufacturing variation caused by the AM process, as larger diameters reliably improved performance across both Nylon PA11 and Aluminium 6063.

A deeper analysis of results with respect to the design of artefacts provides the paper's second contribution. The analysis reveals that the constant-mass global hollowing used in this investigation fails to improve the overall stiffness of complex GD structures, thereby offering no comprehensive lightweighting benefits. Instead, a key insight is that hollow geometry only yields performance gains in long, slender regions constrained by global buckling or flexural stiffness. Conversely, it degrades performance in short, compact joints that rely on shear stiffness and are vulnerable to local shell buckling. Since this investigation focussed on the global post-optimisation hollowing of solid components, two logical future work pathways can be proposed. First, post-process hollowing could be applied selectively, targeting only the long, straight struts while preserving solid geometry at the shear critical joints. Second, natural hole nucleation could be integrated directly into the GD

algorithms. This would enable the optimiser to dynamically evaluate and allocate internal voids during the generation process, providing greater design freedom and therefore structural efficiency than current methodologies.

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